

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Hackathon	Assessment:	Assessment Tool 3– Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and interpret datasets using data preprocessing, statistical analysis, visualization, and machine learning techniques for effective data-driven insights and decision-making.		
Student Name:	Y. Jayesh Ranjan	Reg. No.:	192472201
Section / Batch:	CSE – AI	Date:	29/08/26

Code of Conduct

I Y. Jayesh Ranjan – 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

DSA0405 Fundamentals of Data Science

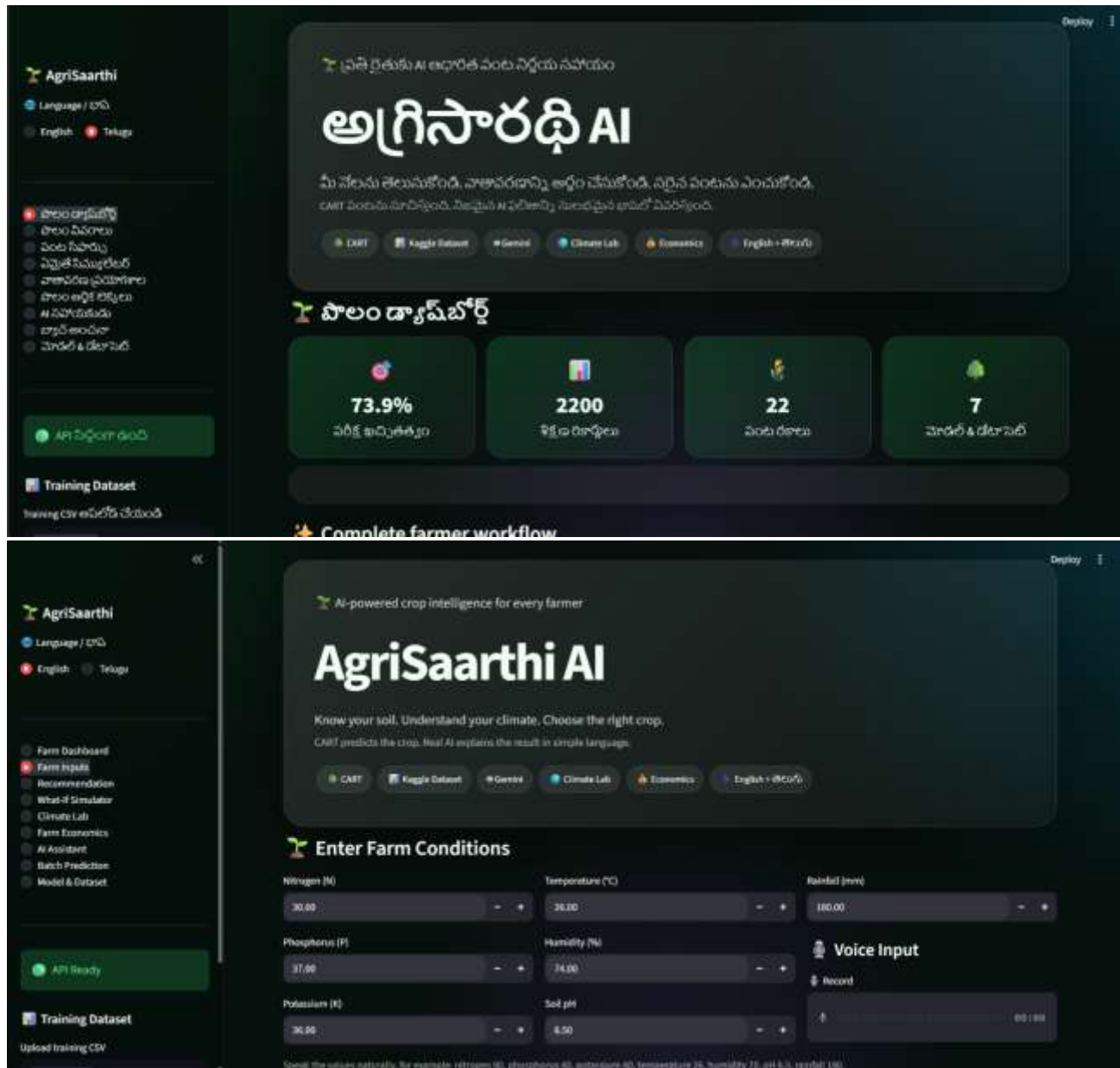
CO5-AT3 – Hackathon

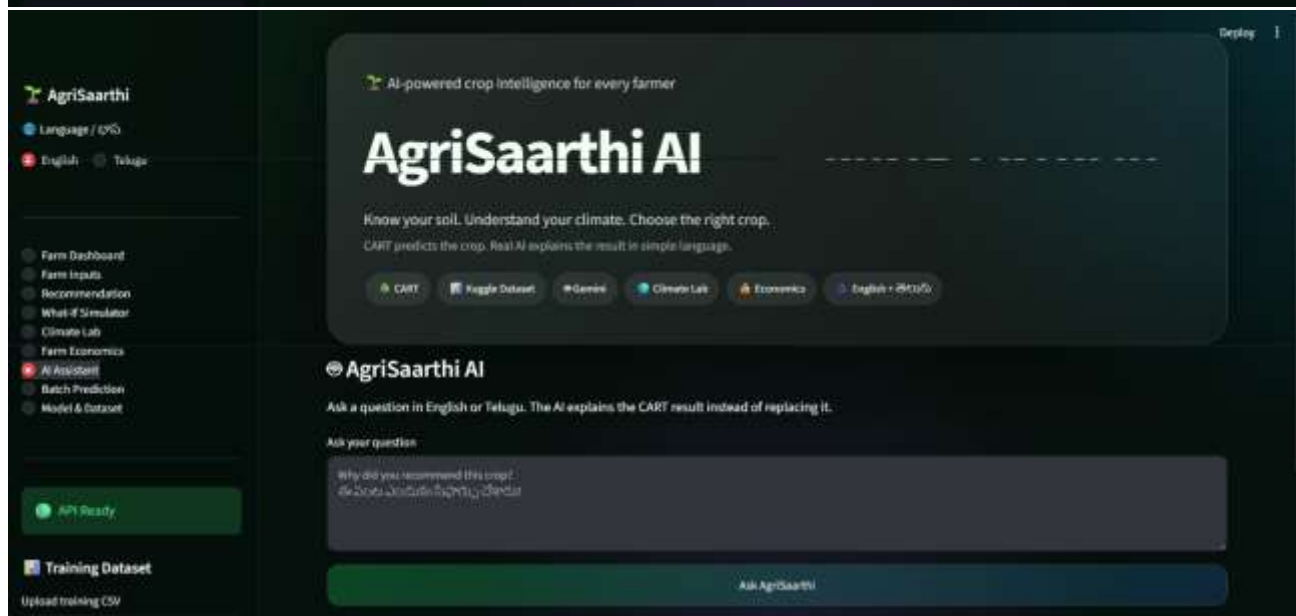
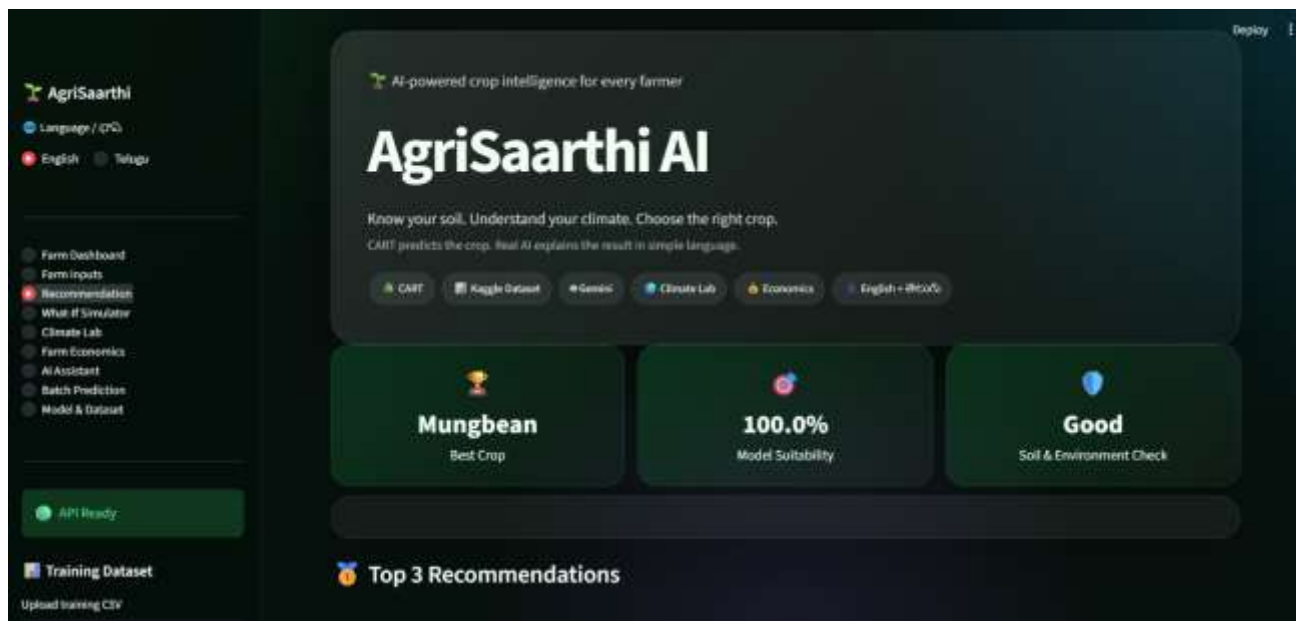
Code:

```
File Edit Selection View Go Run Terminal Help
Crop_recommendation.py x app.py x
app.py
1
2 import os
3 import time
4 import io
5 import numpy as np
6 import pandas as pd
7 import streamlit as st
8 import plotly-express as px
9 from datetime import datetime
10 from sklearn.feature_extraction import DecisionTreeClassifier, export_text
11 from sklearn.model_selection import train_test_split
12 from sklearn.metrics import accuracy_score
13 from google import gmail
14 from google.gmail import types
15
16 load_data()
17
18 st.set_page_config(
19     page_title="AgriSaathi AI",
20     page_icon="🌾",
21     layout="wide",
22     initial_sidebar_state="expanded",
23 )
24
25 # AGRISAATHI AI
26 # the combined project:
27 # * Kaggle/Local dataset training
28 # * Manual Farmer Input
29 # * CAB crop recommendation
30 # * Top-3 recommendations
31 # * Sustainable CAB decision path
32 # * Soil/environment health indicators
33 # * What-If simulator
34 # * Climate resilience scenarios
35 # * Farm economics calculator
```

```
File Edit Selection View Go Run Terminal Help
Crop_recommendation.py x app.py x
app.py
72
73 "button": "Watch Prediction",
74 "technical": "Model & Dataset",
75 "language": "Language",
76 "english": "English",
77 "telugu": "Telugu",
78 "training": "Training Dataset",
79 "upload_train": "Upload training CSV",
80 "farm": "Enter Farm Condition",
81 "n": "Nitrogen (kg)",
82 "p": "Phosphorus (kg)",
83 "k": "Potassium (kg)",
84 "temp": "Temperature (°C)",
85 "humidity": "Humidity (%)",
86 "ph": "Soil pH",
87 "rain": "Rainfall (mm)",
88 "analyse": "Analyse My Farm",
89 "voice": "Voice Input",
90 "voice_help": "Speak the values naturally, for example: nitrogen 40, phosphorus 40, potassium 40, temperature 25, humidity 60",
91 "best": "Best Crop",
92 "confidence": "Model Suitability",
93 "top3": "Top 3 Recommendations",
94 "why": "Why this crop?",
95 "path": "CAB Decision Path",
96 "health": "Soil & Environment Check",
97 "good": "Good",
98 "attention": "Attention",
99 "critical": "Critical",
100 "climate_title": "Climate Resilience Tab",
101 "climate_desc": "Test better and weather/climate scenarios and see whether the recommendation changes.",
102 "whatif_title": "What-If Simulator",
103 "whatif_desc": "Move the sliders and immediately test a different farm condition.",
104 "current": "Current",
105 "scenario": "Scenario",
106 "changed": "Recommendation changed",
107 "stable": "Recommendation stable",
108 "economy_title": "Farm Economics",
109
```

Implementation:







Geotag:

